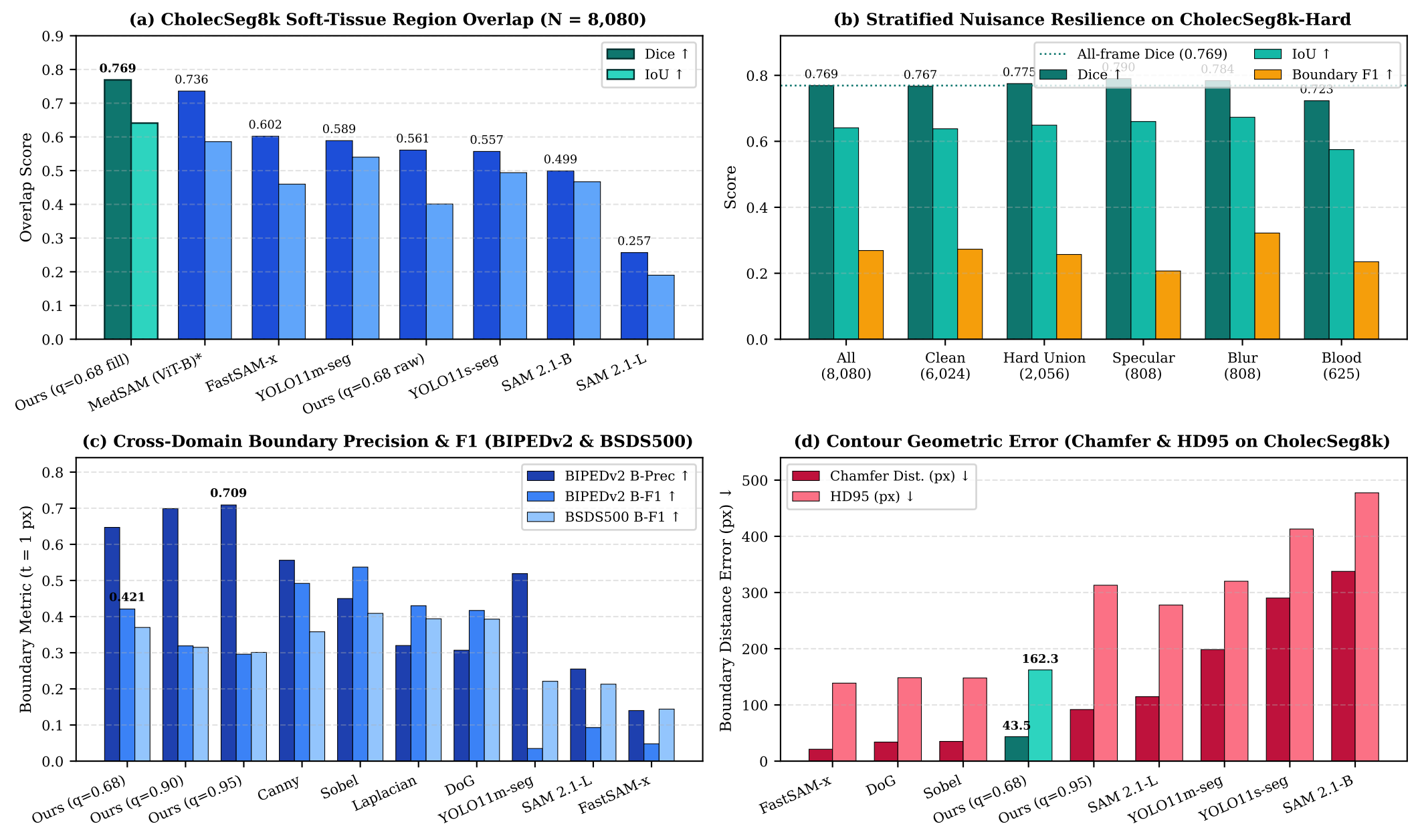
(a) CholecSeg8k Soft-Tissue Region Overlap (N = 8,080)
Overlap Score
Dice ↑
IoU ↑
0.769
0.736
0.602
0.589
0.561
0.557
0.499
0.257
Ours (q=0.68 fill)
MedSAM (ViT-B)*
FastSAM-x
YOLO11m-seg
Ours (q=0.68 raw)
YOLO11s-seg
SAM 2.1-B
SAM 2.1-L
(b) Stratified Nuisance Resilience on CholecSeg8k-Hard
Score
All-frame Dice (0.769)
Dice ↑
IoU ↑
Boundary F1 ↑
0.769
0.767
0.775
0.790
0.784
0.723
All (8,080)
Clean (6,024)
Hard Union (2,056)
Specular (808)
Blur (808)
Blood (625)
(c) Cross-Domain Boundary Precision & F1 (BIPEDv2 & BSDS500)
Boundary Metric (t = 1 px)
BIPEDv2 B-Prec ↑
BIPEDv2 B-F1 ↑
BSDS500 B-F1 ↑
0.421
0.709
Ours (q=0.68)
Ours (q=0.90)
Ours (q=0.95)
Canny
Sobel
Laplacian
DoG
YOLO11m-seg
SAM 2.1-L
FastSAM-x
(d) Contour Geometric Error (Chamfer & HD95 on CholecSeg8k)
Boundary Distance Error (px) ↓
Chamfer Dist. (px) ↓
HD95 (px) ↓
43.5
162.3
FastSAM-x
DoG
Sobel
Ours (q=0.68)
Ours (q=0.95)
SAM 2.1-L
YOLO11m-seg
YOLO11s-seg
SAM 2.1-B